

A Security Strength Statement for Origami Submission

Among the current Origami parameter instances, Origami-128 provides 128-bit classical security and 64-bit quantum security. Its quantum-security bottleneck is the Grover search over the 128-bit master secret expansion seed $C_{\text{seed}}^q = 2^{\kappa_{\text{sk}}/2} = 2^{64}$. According to the algorithm submission requirements, the 128-bit classical security category requires a quantum cryptographic strength of at least 80 bits. Therefore, the current Origami-128 instance does not qualify as a parameter instance satisfying this security category. To achieve an 80-bit quantum cryptographic strength, the master secret expansion seed must satisfy $2^{\kappa_{\text{sk}}/2} \geq 2^{80}$, which gives $\kappa_{\text{sk}} \geq 160$. Hence, the master secret expansion seed must provide at least a 160-bit classical search space so that the corresponding Grover-search complexity reaches 2^{80} . For the current Origami-128 structural parameters $q = 16$ and $(N, M) = (200, 104)$, the quantum target-space claw-finding complexity and the fresh-target preimage-search complexity are approximately $2^{138.7}$ and 2^{208} , respectively, both exceeding 2^{80} . Therefore, increasing the master secret expansion seed length to at least 160 bits removes the current 2^{64} Grover-search bottleneck.

The Origami-128 parameter instance is retained in the main body to provide a complete view of Origami’s scaling behavior across different parameter sizes and to serve as a baseline for analyzing the evolution of key size, signature size, storage overhead, and implementation performance. This instance follows the same algorithmic structure and implementation workflow as the higher-security instances, and retaining its results helps illustrate the impact of parameter growth on performance and resource consumption. Origami-128 is therefore retained only for parameter-scaling and performance evaluation, rather than as a parameter instance satisfying the 128-bit classical / 80-bit quantum security category of the submission requirements. Table 14 summarizes the correspondence between the security strengths of the Origami parameter settings and the submission requirements.

Table 14 – Comparison of Origami security strengths with the submission requirements.

Parameter	Classical Security	Quantum Security	Submission Requirement
Origami-128	128 bits	64 bits	128-bit classical and ≥ 80 -bit quantum
Origami-160	160 bits	80 bits	128-bit classical and ≥ 80 -bit quantum
Origami-256	256 bits	128 bits	256-bit classical and ≥ 128 -bit quantum
Origami-384	384 bits	192 bits	384-bit classical and ≥ 192 -bit quantum
Origami-512	512 bits	256 bits	512-bit classical and ≥ 256 -bit quantum

Origami-256, Origami-384, and Origami-512 provide classical/quantum cryptographic strengths of 256/128 bits, 384/192 bits, and 512/256 bits, respectively, and satisfy the corresponding security-strength requirements of the algorithm submission.